SOFTWARE ENGINEERING

LAB ASSIGNMENT 08

State Transition Diagram

Digital Wallet Management System (DWMS)

SUBMITTED BY — TEAM 5

Suryansh Dwivedi — 2400290100314
Shubh Tiwari — 2400290100296
Subrat Dwivedi — 2400290100311
Tejasvi Gupta — 2400290100322

Department of Computer Science & Engineering

Date: 28 October 2026

Project basis: Team 5 Software Requirements Specification, Version 1.0

1. Introduction to State Transition Diagrams

A UML State Machine Diagram models the lifecycle of an entity through states and transitions. States represent meaningful conditions, while transitions are triggered by events or guarded by conditions. This model is particularly useful for financial transactions because a transaction can be initiated, validated, processed externally, and then reach a terminal outcome.

1.1 Objectives

- Define a transaction lifecycle consistent with the DWMS SRS.
- Show successful, failed, and cancelled outcomes.
- Make clear that wallet balances are updated only after confirmed success and that outcomes remain auditable.

2. Transaction Lifecycle

The SRS requires the system to report payment success/failure, record every payment attempt, and avoid partial wallet updates. The following state model represents a generic transaction used for funding, transfer, or payment. Some states are design-level abstractions that make the required lifecycle explicit; exact status names may be refined during implementation.

UML DIAGRAM

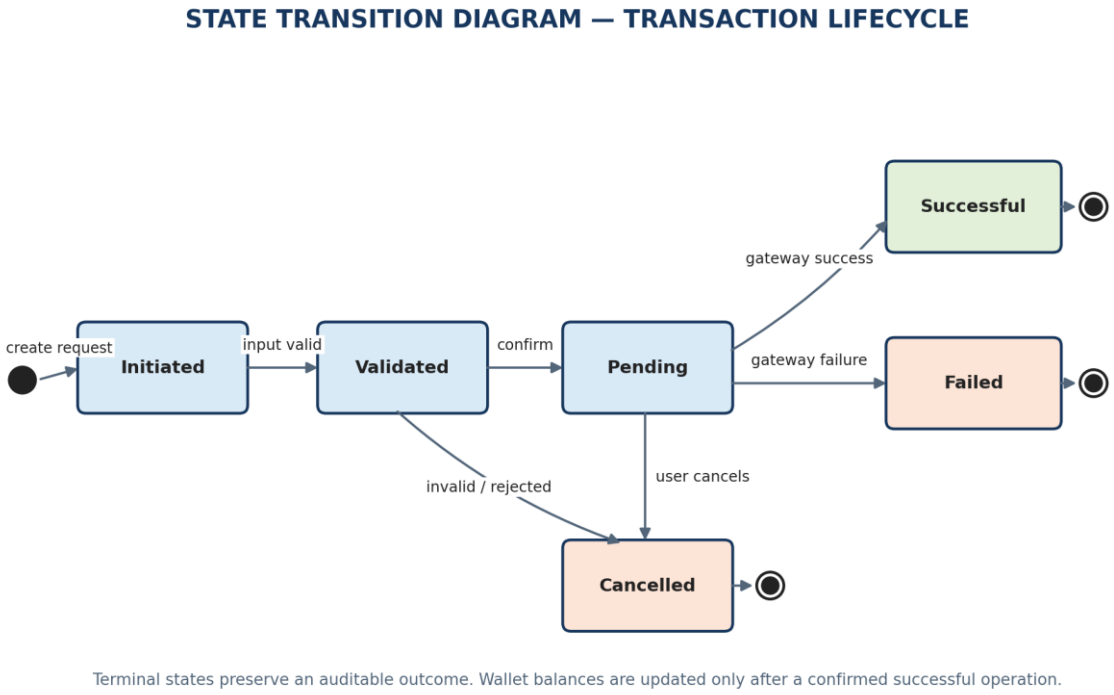


Figure 1. State transition diagram for the transaction lifecycle.

3. State Definitions

- Initiated: a user has submitted a transaction request, but it has not yet passed all checks.
- Validated: the request is valid and required preconditions, such as sufficient balance, have been satisfied.
- Pending: the user has confirmed and the system is awaiting processing/confirmation from the relevant service or commit operation.
- Successful: the operation has completed successfully; the ledger is updated and the user can be notified.
- Failed: processing or validation did not complete successfully. The failure status is recorded and the wallet is not incorrectly credited/debited.
- Cancelled: the user cancels before the operation is committed. The request ends without a financial balance change.

4. Transition Rules

- Initiated → Validated when input and business rules pass.
- Validated → Pending after explicit user confirmation.
- Pending → Successful only after confirmed success and consistent ledger update.
- Pending → Failed when the external service or processing reports failure.
- Initiated/Validated → Cancelled when the request is rejected or cancelled before commitment.

Conclusion

The state transition diagram describes the transaction lifecycle and separates in-progress states from terminal outcomes. It reinforces the SRS requirements for explicit confirmation, final status reporting, auditable transaction records, and balance consistency. The model can guide implementation of status enums and tests for every transition.